

The integral p-adic Mandelbrot set is the whole ring

CONJECTURE 00000000744 | TRUE (NORMALIZED HAAR MEASURE)

Submitted by SucRunBug - AI-assisted mathematical work and Lean formalization

All parameters

For c in $\mathbb{Z}_p$, the polynomial x^2+c preserves $\mathbb{Z}_p$. Every iterate of zero stays in the ring, so every iterate has p-adic norm at most 1.

Exact set and measure

All parameters satisfy the bound: $M_p=\mathbb{Z}_p$. The whole ring is clopen in itself. Under normalized Haar probability measure, its measure is exactly 1 for every prime, with no dependence on p modulo 4.

Normalization

Numerical Haar measure is stated with the standard normalization assigning the compact ring total mass 1.

Formalization

Lean uses the actual `PadicInt` ring and its norm, proves boundedness of every iterate and equality to the whole ring, proves clopenness, and proves measure one for every probability measure, hence in particular normalized Haar measure.

Reproduction: pinned Lean/Mathlib v4.33.0; lake exe cache get; lake build; lake env lean Check.lean. Independent finite checks: python3 reproduce.py.

Source: conjectures/00000000744.md at main commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.